

```
import sys
!{sys.executable} -m pip install transformers[sentencepiece] datasets evaluate accelerate
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: transformers[sentencepiece] in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (24.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (2024.11.23)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.20.1)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.5.1)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (4.67.1)
Requirement already satisfied: sentencepiece!=0.1.92,>=0.1.91 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.2.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (5.29.6)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0) (2024.10.1)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.10.0+cpu)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0) (4.1.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (1.2.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (0.23.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (0.24.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (4.12.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2025.1.1)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (0.3.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (1.20.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (4.7.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (1.0.7)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (0.14.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0) (1.3.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (1.3.0)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (13.9.0)
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM, pipeline
import torch
import transformers # Add this line to import the transformers module
print(f"Transformers version: {transformers.__version__}")
print(f"PyTorch version: {torch.__version__}")
print("Installation and import verified successfully!")
```

```
Transformers version: 5.0.0
PyTorch version: 2.10.0+cpu
Installation and import verified successfully!
```

Start coding or [generate](#) with AI.

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Define the model checkpoint for English to French translation
```

```

model_checkpoint = "Helsinki-NLP/opus-mt-en-fr"

# Explicitly load tokenizer and model
tokenizer = AutoTokenizer.from_pretrained(model_checkpoint)
model = AutoModelForSeq2SeqLM.from_pretrained(model_checkpoint)

# Define a custom translation function to bypass pipeline issues
def translate_text(text):
    inputs = tokenizer(text, return_tensors="pt")
    translated_tokens = model.generate(**inputs, max_length=100)
    translated_text = tokenizer.decode(translated_tokens[0], skip_special_tokens=True)
    return translated_text

print(f"Tokenizer and Model initialized for: {model_checkpoint}")

/usr/local/lib/python3.12/dist-packages/transformers/models/ Marian/tokenization_marian.py:176: UserWarning: Recommended: pip ins
warnings.warn("Recommended: pip install sacremoses.")

Loading weights: 100% 258/258 [00:00<00:00, 422.11it/s, Materializing param=model.shared.weight]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, bu
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, bu
Tokenizer and Model initialized for: Helsinki-NLP/opus-mt-en-fr

```

```

text_to_translate = "Hello, how are you today?"
# Call the custom translate_text function instead of the pipeline object
translation_result_text = translate_text(text_to_translate)

print(f"Original text: {text_to_translate}")
print(f"Translated text: {translation_result_text}")

if translation_result_text:
    print("Model loaded and verified successfully with a sample translation!")
else:
    print("Model verification failed.")

```

Original text: Hello, how are you today?
Translated text: Bonjour, comment allez-vous aujourd'hui ?
Model loaded and verified successfully with a sample translation!

```

# Students must input their prepared dataset sentences here
input_sentences = [
    "The quick brown fox jumps over the lazy dog.", # Simple declarative
    "How do you say 'hello' in French?", # Interrogative
    "Please close the door quietly.", # Imperative
    "If it rains tomorrow, we will stay inside.", # Conditional
    "The book was written by a famous author.", # Passive voice
    "It's raining cats and dogs today!", # Idiomatic expression
    "She believes that the project, which was started last year and faced numerous delays, will finally be completed next month",
    "The old man and his dog walked slowly down the long, winding road toward the village nestled in the valley.", # Longer, de
    "I went to the store, bought some apples, and came home.", # Compound sentence
    "Do you know where my keys are? I can't find them anywhere!" # Multi-sentence query
]

print(f"Prepared {len(input_sentences)} sentences for translation.")

```

Prepared 10 sentences for translation.

```

# Process and translate the input sentences
print("\n--- Translation Results ---\n")
for i, sentence in enumerate(input_sentences):
    translated_sentence = translate_text(sentence)
    print(f"Original {i+1}: {sentence}")
    print(f"Translated {i+1}: {translated_sentence}\n")

print("--- Translation process complete --- ")

```

--- Translation Results ---

Original 1: The quick brown fox jumps over the lazy dog.
Translated 1: Le renard brun rapide saute sur le chien paresseux.

Original 2: How do you say 'hello' in French?
Translated 2: Comment tu dis "bonjour" en français ?

Original 3: Please close the door quietly.

Translated 3: S'il vous plaît fermez la porte tranquillement.

Original 4: If it rains tomorrow, we will stay inside.

Translated 4: S'il pleut demain, nous resterons à l'intérieur.

Original 5: The book was written by a famous author.

Translated 5: Le livre a été écrit par un auteur célèbre.

Original 6: It's raining cats and dogs today!

Translated 6: Il pleut des chats et des chiens aujourd'hui !

Original 7: She believes that the project, which was started last year and faced numerous delays, will finally be completed next

Translated 7: Elle estime que le projet, qui a débuté l'année dernière et a connu de nombreux retards, sera finalement achevé le

Original 8: The old man and his dog walked slowly down the long, winding road toward the village nestled in the valley.

Translated 8: Le vieil homme et son chien descendirent lentement la longue route sinueuse vers le village niché dans la vallée.

Original 9: I went to the store, bought some apples, and came home.

Translated 9: Je suis allé au magasin, j'ai acheté des pommes, et je suis rentré à la maison.

Original 10: Do you know where my keys are? I can't find them anywhere!

Translated 10: Tu sais où sont mes clés ?

--- Translation process complete ---

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
```

```
# Define the model checkpoint for English to German translation
```

```
model_checkpoint_de = "Helsinki-NLP/opus-mt-en-de"
```

```
# Explicitly load tokenizer and model for German
```

```
tokenizer_de = AutoTokenizer.from_pretrained(model_checkpoint_de)
```

```
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_checkpoint_de)
```

```
def translate_text_de(text):
```

```
    inputs = tokenizer_de(text, return_tensors="pt")
```

```
    translated_tokens = model_de.generate(**inputs, max_length=100)
```

```
    translated_text = tokenizer_de.decode(translated_tokens[0], skip_special_tokens=True)
```

```
    return translated_text
```

```
print(f"Tokenizer and Model initialized for: {model_checkpoint_de}")
```

```
config.json:      1.33k/? [00:00<00:00, 28.3kB/s]
```

```
tokenizer_config.json: 100%                               42.0/42.0 [00:00<00:00, 1.25kB/s]
```

```
source.spm: 100%                                         768k/768k [00:00<00:00, 11.0MB/s]
```

```
target.spm: 100%                                         797k/797k [00:00<00:00, 22.3MB/s]
```

```
vocab.json:      1.27M/? [00:00<00:00, 25.3MB/s]
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/arian/tokenization_arian.py:176: UserWarning: Recommended: pip install sacremoses.")
```

```
pytorch_model.bin: 100%                                298M/298M [00:02<00:00, 274MB/s]
```

```
Loading weights: 100%                                258/258 [00:00<00:00, 468.95it/s, Materializing param=model.shared.weight]
```

```
model.safetensors: 100%                                298M/298M [00:03<00:00, 149MB/s]
```

```
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but
```

```
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but
```

```
generation_config.json: 100%                            293/293 [00:00<00:00, 16.3kB/s]
```

```
Tokenizer and Model initialized for: Helsinki-NLP/opus-mt-en-de
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
```

```
# Define the model checkpoint for English to Hindi translation
```

```
model_checkpoint_hi = "Helsinki-NLP/opus-mt-en-hi"
```

```
# Explicitly load tokenizer and model for Hindi
```

```
tokenizer_hi = AutoTokenizer.from_pretrained(model_checkpoint_hi)
```

```
model_hi = AutoModelForSeq2SeqLM.from_pretrained(model_checkpoint_hi)
```

```
def translate_text_hi(text):
```

```
    inputs = tokenizer_hi(text, return_tensors="pt")
```

```
    translated_tokens = model_hi.generate(**inputs, max_length=100)
```

```
    translated_text = tokenizer_hi.decode(translated_tokens[0], skip_special_tokens=True)
```

```
return translated_text

print(f"Tokenizer and Model initialized for: {model_checkpoint_hi}")

config.json:      1.39k/? [00:00<00:00, 83.7kB/s]

tokenizer_config.json: 100%                               44.0/44.0 [00:00<00:00, 3.50kB/s]

source.spm: 100%                                         812k/812k [00:00<00:00, 40.1MB/s]

target.spm: 100%                                         1.07M/1.07M [00:00<00:00, 42.8MB/s]

vocab.json:      2.10M/? [00:00<00:00, 49.2MB/s]

pytorch_model.bin: 100%                                306M/306M [00:03<00:00, 159MB/s]

model.safetensors: 100%                                306M/306M [00:04<00:00, 103MB/s]

Loading weights: 100%                                258/258 [00:01<00:00, 161.78it/s, Materializing param=model.shared.weight]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but the config does not contain the key 'decoder.embed_tokens.weight'.
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but the config does not contain the key 'encoder.embed_tokens.weight'.

generation_config.json: 100%                             293/293 [00:00<00:00, 7.29kB/s]

Tokenizer and Model initialized for: Helsinki-NLP/opus-mt-en-hi
```

```
print("\n--- German Translation Results ---\n")
for i, sentence in enumerate(input_sentences):
    translated_sentence_de = translate_text_de(sentence)
    print(f"Original {i+1}: {sentence}")
    print(f"Translated (DE) {i+1}: {translated_sentence_de}\n")

print("\n--- Hindi Translation Results ---\n")
for i, sentence in enumerate(input_sentences):
    translated_sentence_hi = translate_text_hi(sentence)
    print(f"Original {i+1}: {sentence}")
    print(f"Translated (HI) {i+1}: {translated_sentence_hi}\n")

print("--- Multilingual translation process complete ---")
```

```
--- German Translation Results ---

Original 1: The quick brown fox jumps over the lazy dog.
Translated (DE) 1: Der schnelle Braunfuchs springt über den faulen Hund.

Original 2: How do you say 'hello' in French?
Translated (DE) 2: Wie sagt man "Hallo" auf Französisch?

Original 3: Please close the door quietly.
Translated (DE) 3: Bitte schließen Sie die Tür leise.

Original 4: If it rains tomorrow, we will stay inside.
Translated (DE) 4: Wenn es morgen regnet, bleiben wir drinnen.

Original 5: The book was written by a famous author.
Translated (DE) 5: Das Buch wurde von einem berühmten Autor geschrieben.

Original 6: It's raining cats and dogs today!
Translated (DE) 6: Es regnet heute Katzen und Hunde!

Original 7: She believes that the project, which was started last year and faced numerous delays, will finally be completed next year.
Translated (DE) 7: Sie glaubt, dass das Projekt, das im vergangenen Jahr gestartet wurde und mit zahlreichen Verzögerungen konfrontiert war, endlich im nächsten Jahr fertiggestellt werden wird.

Original 8: The old man and his dog walked slowly down the long, winding road toward the village nestled in the valley.
Translated (DE) 8: Der alte Mann und sein Hund gingen langsam die lange, kurvenreiche Straße hinunter in Richtung des Dorfes, das im Tal liegt.

Original 9: I went to the store, bought some apples, and came home.
Translated (DE) 9: Ich ging in den Laden, kaufte Äpfel und kam nach Hause.

Original 10: Do you know where my keys are? I can't find them anywhere!
Translated (DE) 10: Weißt du, wo meine Schlüssel sind?

--- Hindi Translation Results ---

Original 1: The quick brown fox jumps over the lazy dog.
Translated (HI) 1: कम से कम भूरे रंग का कपड़ा आलसी कुत्ते पर कूदता है।

Original 2: How do you say 'hello' in French?
Translated (HI) 2: आप फ्रेंच में 'हैग' कैसे कहते हैं?
```

Original 3: Please close the door quietly.
Translated (HI) 3: कृपया दरवाजा चुपचाप बंद करें.

Original 4: If it rains tomorrow, we will stay inside.
Translated (HI) 4: यह कल बारिश होती है, तो हम अंदर ही रहेंगे.

Original 5: The book was written by a famous author.
Translated (HI) 5: इस किताब को एक जाने - माने लेखक ने लिखा था ।

Original 6: It's raining cats and dogs today!
Translated (HI) 6: यह आजबकियां और कुत्तों की बारिश कर रहा है!

Original 7: She believes that the project, which was started last year and faced numerous delays, will finally be completed ne:
Translated (HI) 7: वह विश्वास करती है कि वह परियोजना, जो पिछले साल शुरू की गयी थी और अनेक देरी का सामना किया जाएगा, अंततः अगले महीने पूरी

```
# Define a new set of complex input sentences for testing
```

```
new_input_sentences = [
    "Despite the persistent rain, which threatened to dampen spirits, the festival attendees continued to dance and celebrate w
    "The scientist meticulously analyzed the data, and she discovered a previously unknown anomaly, so she decided to conduct f
    "If you finish your work early, you can go home, or you can assist your colleagues with their tasks.", # Complex with optic
    "Could you please explain the theoretical underpinnings of recurrent neural networks, focusing on their application in natu
    "The economic forecast, which was released yesterday, indicated a potential downturn in the market, leading to widespread c
    "While the cat was sleeping soundly on the armchair, the mischievous puppy quietly stole a cookie from the kitchen counter.
    "I enjoy reading historical novels, but my brother prefers science fiction, and our sister is fond of fantasy literature.",
    "To ensure the project's success, we must allocate sufficient resources, establish clear communication channels, and define
    "Before you embark on your journey, remember to pack warm clothes, check your vehicle's oil, and confirm your accommodation
    "How does the attention mechanism in Transformer models improve performance compared to traditional sequence-to-sequence ar
]
```

```
print(f"Prepared {len(new_input_sentences)} new complex sentences for translation.")
```

Prepared 10 new complex sentences for translation.

```
print("\n--- French Translation Results (Complex Sentences) ---\n")
for i, sentence in enumerate(new_input_sentences):
    translated_sentence = translate_text(sentence) # Using the existing English-French translator
    print(f"Original {i+1}: {sentence}")
    print(f"Translated (FR) {i+1}: {translated_sentence}\n")
```

--- French Translation Results (Complex Sentences) ---

Original 1: Despite the persistent rain, which threatened to dampen spirits, the festival attendees continued to dance and celeb
Translated (FR) 1: Malgré la pluie persistante, qui menaçait d'apaiser les esprits, les participants au festival continuèrent à

Original 2: The scientist meticulously analyzed the data, and she discovered a previously unknown anomaly, so she decided to con
Translated (FR) 2: La scientifique a analysé méticuleusement les données, et elle a découvert une anomalie précédemment inconnue

Original 3: If you finish your work early, you can go home, or you can assist your colleagues with their tasks.
Translated (FR) 3: Si vous terminez votre travail tôt, vous pouvez rentrer chez vous ou aider vos collègues dans leurs tâches.

Original 4: Could you please explain the theoretical underpinnings of recurrent neural networks, focusing on their application i
Translated (FR) 4: Pourriez-vous expliquer les fondements théoriques des réseaux neuronaux récurrents, en mettant l'accent sur l

Original 5: The economic forecast, which was released yesterday, indicated a potential downturn in the market, leading to widesp
Translated (FR) 5: Les prévisions économiques, qui ont été publiées hier, ont révélé un ralentissement potentiel du marché, ce q

Original 6: While the cat was sleeping soundly on the armchair, the mischievous puppy quietly stole a cookie from the kitchen co
Translated (FR) 6: Pendant que le chat dormait bien sur le fauteuil, le chiot malicieux a discrètement volé un cookie au comptoi

Original 7: I enjoy reading historical novels, but my brother prefers science fiction, and our sister is fond of fantasy literat
Translated (FR) 7: J'aime lire des romans historiques, mais mon frère préfère la science-fiction, et notre sœur aime la littérat

Original 8: To ensure the project's success, we must allocate sufficient resources, establish clear communication channels, and
Translated (FR) 8: Pour assurer le succès du projet, nous devons allouer des ressources suffisantes, établir des canaux de commu

Original 9: Before you embark on your journey, remember to pack warm clothes, check your vehicle's oil, and confirm your accommo
Translated (FR) 9: Avant d'embarquer dans votre voyage, n'oubliez pas d'emballer des vêtements chauds, de vérifier l'huile de vo

Original 10: How does the attention mechanism in Transformer models improve performance compared to traditional sequence-to-sequ
Translated (FR) 10: Comment le mécanisme d'attention dans les modèles Transformer améliore-t-il les performances par rapport aux

```
print("\n--- German Translation Results (Complex Sentences) ---\n")
for i, sentence in enumerate(new_input_sentences):
    translated_sentence_de = translate_text_de(sentence) # Using the existing English-German translator
    print(f"Original {i+1}: {sentence}")
    print(f"Translated (DE) {i+1}: {translated_sentence_de}\n")
```

--- German Translation Results (Complex Sentences) ---

Original 1: Despite the persistent rain, which threatened to dampen spirits, the festival attendees continued to dance and celeb
Translated (DE) 1: Trotz des anhaltenden Regens, der zu dämpfen drohte, tanzten und feierten die Festivalteilnehmer mit unerschü

Original 2: The scientist meticulously analyzed the data, and she discovered a previously unknown anomaly, so she decided to con
Translated (DE) 2: Die Wissenschaftlerin analysierte die Daten sorgfältig und entdeckte eine bisher unbekannte Anomalie, so dass

Original 3: If you finish your work early, you can go home, or you can assist your colleagues with their tasks.

Translated (DE) 3: Wenn Sie Ihre Arbeit vorzeitig beenden, können Sie nach Hause gehen, oder Sie können Ihre Kollegen bei ihren

Original 4: Could you please explain the theoretical underpinnings of recurrent neural networks, focusing on their application i
Translated (DE) 4: Könnten Sie bitte die theoretischen Grundlagen rezidivierender neuronaler Netzwerke erklären, die sich auf ih

Original 5: The economic forecast, which was released yesterday, indicated a potential downturn in the market, leading to widesp
Translated (DE) 5: Die Wirtschaftsprognose, die gestern veröffentlicht wurde, deutete auf einen möglichen Marktabschwung hin, de

Original 6: While the cat was sleeping soundly on the armchair, the mischievous puppy quietly stole a cookie from the kitchen co
Translated (DE) 6: Während die Katze laut auf dem Sessel schlief, stahl der schelmische Welpe leise einen Keks von der Küchenthe

Original 7: I enjoy reading historical novels, but my brother prefers science fiction, and our sister is fond of fantasy literat
Translated (DE) 7: Ich lese gern historische Romane, aber mein Bruder bevorzugt Science-Fiction, und unsere Schwester liebt Fant

Original 8: To ensure the project's success, we must allocate sufficient resources, establish clear communication channels, and
Translated (DE) 8: Um den Erfolg des Projekts zu gewährleisten, müssen wir ausreichende Ressourcen bereitstellen, klare Kommunik

Original 9: Before you embark on your journey, remember to pack warm clothes, check your vehicle's oil, and confirm your accommo
Translated (DE) 9: Bevor Sie Ihre Reise beginnen, denken Sie daran, warme Kleidung zu packen, das Öl Ihres Fahrzeugs zu überprüf

Original 10: How does the attention mechanism in Transformer models improve performance compared to traditional sequence-to-sequ
Translated (DE) 10: Wie verbessert der Aufmerksamkeitsmechanismus in Transformer-Modellen die Leistungsfähigkeit im Vergleich zu

```
print("\n--- Hindi Translation Results (Complex Sentences) ---\n")
for i, sentence in enumerate(new_input_sentences):
    translated_sentence_hi = translate_text_hi(sentence) # Using the existing English-Hindi translator
    print(f"Original {i+1}: {sentence}")
    print(f"Translated (HI) {i+1}: {translated_sentence_hi}\n")
```

```
print("--- Complex sentence translation process complete ---")
```

--- Hindi Translation Results (Complex Sentences) ---

Original 1: Despite the persistent rain, which threatened to dampen spirits, the festival attendees continued to dance and celeb
Translated (HI) 1: इसके बावजूद, जब लोग लगातार बारिश करते रहे, तो लोगों ने जोश के साथ नाच - गाना जारी रखा और पूरे जोश के साथ जश्न मनाया ।

Original 2: The scientist meticulously analyzed the data, and she discovered a previously unknown anomaly, so she decided to con
Translated (HI) 2: वैज्ञानिक ने पूरी तरह से डेटा की जाँच की, और उसने एक पहले एक अनजान अनजानी खोज की, सो उसने और अधिक परीक्षण करने का फै

Original 3: If you finish your work early, you can go home, or you can assist your colleagues with their tasks.

Translated (HI) 3: अगर आप अपना काम जल्दी खत्म कर लेते हैं, तो आप घर जा सकते हैं या आप अपने साथ काम करनेवालों की मदद कर सकते हैं ।

Original 4: Could you please explain the theoretical underpinnings of recurrent neural networks, focusing on their application i
Translated (HI) 4: क्या आप यह समझा सकते हैं कि तंत्रिकाओं की तंत्रिकाओं की स्वाभाविक भाषा में उनकी अनुप्रयोग पर ध्यान केंद्रित करते हुए, क्या आप कृपया

Original 5: The economic forecast, which was released yesterday, indicated a potential downturn in the market, leading to widesp
Translated (HI) 5: आर्थिक पूर्वानुमान, जो कल रिहा किया गया था, बाज़ार में एक संभावना सामने आयी, और निवेशकों के बीच व्यापक चिंता की ओर ले गया ।

Original 6: While the cat was sleeping soundly on the armchair, the mischievous puppy quietly stole a cookie from the kitchen co
Translated (HI) 6: जबकि बिल्ली तेज़ नींद में सो रहा था, दुष्ट पपी ने चुपचाप रसोई - विरोधी से एक कुकी चुरा लिया ।

Original 7: I enjoy reading historical novels, but my brother prefers science fiction, and our sister is fond of fantasy literat
Translated (HI) 7: मुझे ऐतिहासिक उपन्यास पढ़ने में मज़ा आता है, लेकिन मेरा भाई विज्ञान - कथा पसंद करता है, और हमारी बहन कल्पना - शक्ति के साहित्य

Original 8: To ensure the project's success, we must allocate sufficient resources, establish clear communication channels, and
Translated (HI) 8: परियोजना की सफलता सुनिश्चित करने के लिए, हम पर्याप्त संसाधन वितरित करना चाहिए, स्पष्ट संचार चैनल स्थापित करें, और मुझे परिभाषित

Original 9: Before you embark on your journey, remember to pack warm clothes, check your vehicle's oil, and confirm your accommo
Translated (HI) 9: आने से पहले, गर्म कपड़े धोने के लिए याद रखें, अपने वाहन के तेल की जाँच करें, और अपनी स्टिकिंग बुकिंग की पुष्टि करें.

Original 10: How does the attention mechanism in Transformer models improve performance compared to traditional sequence-to-sequ
Translated (HI) 10: रूपांतरण मॉडलों का ध्यान लगाने से पारंपरिक अनुक्रम-से-से- टिकित रचना की तुलना में कैसे सुधार होता है, विशेष रूप से लंबी निर्भरता

--- Complex sentence translation process complete ---

